ZIMBABWE INTEGRATING CAPITAL AND RISK PROGRAMME

('ZICARP')

TS 3 – Technical Specification for Valuation of Technical Provisions

Version 3

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1. Introduction

1.1 TS3 outlines the liability valuation guidelines that insurers must adhere to when valuing their technical provisions.

1.2 The reliability of technical provisions calculation is the basis for assessing the solvency position of an insurance company. A strong assessment of these technical provisions is therefore required to ensure accurate reporting on the solvency position.

1.3 Valuation of technical provisions is critical in a supervision context. This will give the Insurance and Pensions Commission ('IPEC' or the Commission) a true and fair view of the Insurance Sector in Zimbabwe, which is the basis of the Risk Based Capital (RBC) framework.

1.4 These valuation guidelines will replace liabilities valuation regulations.

Application

1.5 This TS applies to all insurers that are licensed and supervised by the Insurance and Pensions Commission ('IPEC' or the 'Commission').

Effective Date

1.6 This updated Specification Version 3 is effective 1 January 2023 and should be read in conjunction with Circular 23 of 2021 which implements ZICARP.

2. Duties and Accountabilities under ZICARP

2.1. Ultimate responsibility for the solvency assessment and management of the insurer rests with the Board of Directors ('Board'). The Board of an insurance company must ensure that the insurer always meets all the risk-based capital requirements.

2.2. The Actuarial Function will report to the Board and provide assurance that the calculations of solvency capital requirements are accurate and compliant with the requirements of the ZICARP framework.

2.3. The External Auditor must audit the financial health of an insurer in accordance with its legal and regulatory obligations. The Auditor must report to the Board of Directors and IPEC any matters identified during the audit process that may cause the insurer not to be financially sound.

2.4. The Internal Audit function shall perform regular independent and objective assessments of key components of the ZICARP framework.

2.5. The Compliance Function shall have the authority and obligation to promptly inform the chairperson of the board directly in the event of any contravention of the Insurance Act [Chapter 24:07] and any other major non-compliance by a member of management or any staff member for that matter or a material non-compliance by the insurer with the ZICARP framework or an external obligation.

2.6. Detailed roles and accountabilities under ZICARP are further outlined in TS 1 and Governance and Risk Specifications 2, 3 and 4.

3. Technical Provisions Valuation Guidelines

3.1 The following is the recommended framework for the valuation of technical provisions under ZICARP. The framework has been segmented into two:

3.1.1 Non-life Liability Valuation Guidelines for Technical Liabilities; and

3.1.2 Life/Funeral Assurance Liability Valuation Guidelines for Technical Liabilities.

3.2 The technical provisions valuation guidelines have been structured to cover the treatment of the following key components under both short-term and long-term insurance businesses:

3.2.1 Data quality;

3.2.2 Segmentation/ Risk Classification;

3.2.3 Methodology for Valuing Technical Liabilities;

3.2.4 Recognition of Insurance Contracts;

3.2.5 Contract Boundaries;

3.2.6 Recognition of Insurance Premiums;

3.2.7 Valuation of Options and Guarantees;

3.2.8 Treatment of Future Discretionary Benefits;

3.2.9 Recoverables;

3.2.10 Discount Rates; and

3.2.11 Risk Margins.

Cardinal Valuation Principles

3.3 Insurance and reinsurance (insurers) undertakings should ensure that data used in the calculation of technical provisions cover a sufficiently large period of observations that characterise the reality being measured.

3.4 The valuation of technical provisions should allow for the **time value of money** and a criterion for the determination of appropriate rates to be used in the discounting of technical provisions.

3.5 Insurers must segment their insurance obligations into homogeneous risk groups when calculating the value of technical provisions.

3.6 The technical provisions shall be the sum of the Best Estimate Liability and the Risk Margin i.e., the valuation of the technical provisions should incorporate a best estimate and a risk margin, although, under certain conditions, the valuation may be performed as a whole.

3.7 The Best Estimate Liability must be calculated on a gross basis, without deducting the amounts recoverable from reinsurance contracts and other risk mitigation instruments.

3.8 The Best Estimate Liability should correspond to the probability-weighted average of future cash flows taking into account the time value of money.

3.9 Insurers must use actuarial and statistical techniques for the calculation of the best estimate which appropriately reflect the risks that affect the cash flows. These actuarial and statistical techniques should be **proportionate** to the nature, scale and complexity of the insurer's obligations.

3.10 Insurers must establish technical provisions that correspond to the current value of their insurance obligations i.e., the technical provisions shall correspond to the current amount an insurer may pay if the insurer were to transfer the insurer's insurance obligations immediately to another insurer.

3.11 When an insurer values its technical provisions, no subsequent adjustment shall take account of the insurer's credit standing.

3.12 Amounts relating to recoverables must be calculated separately as part of the valuation of technical provisions.

3.13 Insurers are permitted to apply simplified methods in valuing technical provisions, subject to such methods satisfying the principle of proportionality.

4. General Valuation Principles

4.1 The following valuation principles apply to both entities in life and non-life insurance:

Data Quality: Completeness of data, Appropriateness of data & Data checks

4.2 The actuary should consider whether sufficient and reliable data are available to perform actuarial services. Data are sufficient if they include the appropriate information for the work. Data are reliable if they are substantially accurate.

4.3 Insurers and reinsurers should ensure that data relating to different periods are used consistently.

4.4 If necessary, undertakings should apply adjustments to historical data to increase its credibility or enhance its quality as an input to determine more reliable estimates of technical provisions and to better align it with the characteristics of the portfolio being valued and with the future expected development of risks.

4.5 Insurance and reinsurance undertakings should ensure that the actuarial function assesses the accuracy and completeness of data through a sufficiently comprehensive series of checks that allow the detection of any relevant shortcomings.

4.6 Insurance and reinsurance undertakings should ensure that the actuarial function takes into account the relationship between the conclusions of the analysis of the data quality and the selection of the methodologies to be applied to value the technical provisions.

4.7 Insurers should ensure that the actuarial function analyses the adequacy of data to support the assumptions underlying the methodologies to be applied to value the technical provisions. If data does not adequately support the methodologies, then the undertaking should select an alternative methodology.

4.8 In the assessment of completeness of data, undertakings should ensure that the actuarial function considers whether the number of observations and granularity of available data is sufficient and adequate to meet the input requirements for the application of the methodology.

4.9 Insurers and reinsurers should ensure that the actuarial function documents data limitations, including at least:

4.9.1 A description of the shortcomings comprising its causes and any references to other documents where they were identified;

4.9.2 A summary explanation on the impact of the shortcomings in the scope of the calculation of technical provisions regarding its materiality and how it affects this process;

4.9.3 A description of the actions taken by the actuarial function to detect the shortcomings in data, including whether data was complemented or not with other sources and documents; and

4.9.4 A description of how such situations can be remedied in a short term for the intended purpose and any relevant recommendations to be applied to enhance data quality in the future.

Segmentation

4.10 In valuing technical provisions, insurers must segment their insurance obligations into, at a minimum, the sublines of business prescribed in TS 3- A1 of this Guideline.

4.11 Insurers should consider both the nature of the underlying risks being evaluated together and the quality of data in selecting the homogeneous risk groups for the calculations of relevant provisions.

4.12 Insurers should calculate technical provisions using homogeneous risk groups in order to derive assumptions. A homogeneous risk group encompasses a collection of policies with similar risk characteristics.

4.13 In selecting a homogeneous risk group, insurers and reinsurers should achieve an appropriate balance between the credibility of available data, to enable reliable statistical analyses to be performed and the homogeneity of risk characteristics within the group.

4.14 Insurers should define homogeneous risk groups in such a manner that those are expected to be reasonably stable over time. The purpose of segmentation is to prevent potential distortions in the valuation process that may arise from combining dissimilar sublines of business.

4.15 Insurers should ensure consistency between the homogeneous risk groups it uses to assess its gross of reinsurance technical provisions and its reinsurance recoverables.

4.16 Segmentation must be applied to the valuation of the best estimate and the risk margin. Segmentation must also be applied where technical provisions are calculated as a whole.

4.17 For financial soundness purposes, segmentation must be based on the nature of the risks underlying the insurance obligation rather than the legal form of the insurance contract, i.e., the principle of "substance over form".

4.18 Insurers must apply the principle of substance over form when segmenting between life and non-life insurance obligations in the valuation of technical provisions. This requires:

4.18.1 Insurance obligations pursued on a similar technical basis to that of life insurance to be considered as life insurance obligations, even if they are non-life insurance from a legal perspective;

4.18.2 Insurance obligations not pursued on a similar technical basis to that of life insurance to be considered as non-life insurance obligations, even if they are life insurance from a legal perspective; and

4.18.3 Annuities stemming from non-life insurance contracts to be treated as life insurance obligations.

4.19 Where an insurance policy includes life and non-life insurance obligations, the policy must be unbundled into the life and non-life parts wherever possible, and where both parts are material. Where only one of the risks covered by an insurance policy is material, the contract may be bundled and allocated to the main risk.

4.20 Segmentation must be applied to both the valuation of the best estimate and risk margin. Segmentation must also be applied where technical provisions are calculated as a whole.

Contract Boundary

4.21 The Best Estimate Liability must be determined as the discounted value of projected cash flows up to the contract boundary.

4.22 To determine which insurance obligations arise in relation to an insurance contract, the contract boundary is defined as the date on which the insurer has the unilateral right to:

4.22.1 Terminate the contract;

4.22.2 Reject the premiums payable under the contract; or

4.22.3 Amend the premiums or benefits payable under the contract at a future date in such a way that the premiums fully reflect the risks.

4.23 Any insurance obligation provided by the insurer after the contract boundary does not belong to the existing contract unless the insurer can compel the policyholder to pay the premium for those obligations.

4.24 An insurer has a unilateral right to review policy conditions when there are no constraints to prevent it from setting the same price as it would for a new contract, or if it can amend the benefits to be consistent with those that it would provide for the price that it will charge in the future. The unilateral right to review policy conditions should not only reflect a legal right but should also consider policyholders' reasonable expectations, behaviour, as well as market pressures.

4.25 An insurer shall take into account future premiums up to the contract boundary which shall be the point at which an insurer can unilaterally terminate an insurance contract, refuse to accept a premium, vary the premium or the benefits in such a way that the premium fully reflects the risks insured.

4.26 The projection time horizon used in the calculation of the best estimate liability by an insurer or reinsurer shall cover the full lifetime of all the cash inflows (excluding investment returns, interest and dividends) and cash outflows required to settle the obligations related to existing insurance and reinsurance contracts on the date of the valuation.

Recognition of Insurance Contracts

4.27 The valuation of the best estimate must only include future cash flows associated with existing insurance policies.

4.28 Insurers must not recognise, as an asset or a liability, any amounts relating to expected premiums that are outside the contract boundary.

4.29 An insurance policy should be derecognised as an existing contract only when the obligation specified in the policy expires or has been discharged or cancelled.

Recognition of Insurance Premiums

4.30 Non-life insurers must recognise the full premium due or expected to become due, upfront and create a premium debtor equal to the amount expected to become due in future.

4.31 Reinsurers must determine the contract boundaries of their reinsurance contracts taking into consideration when it would be practically possible to cancel a contract. All underlying policies incepting during the contract boundary period should be allowed.

4.32 Insurers that issue policies that provide the policyholder with the option to extend the policy, or to increase the amount covered by simply notifying the insurer, must not recognise the future expected premiums related to such extensions. Such premiums, together with any associated claims and other cash-flows, should be included in the future cash-flow projections. These policies typically apply in non-life insurance, such as engineering guarantees that provide the policyholder with the option to extend the policy at the end of the original guarantee period.

4.33 Life insurance policies involving multiple premiums, where the contract boundary is longer than the intervals between premium payments, must only recognise premiums that have become due prior to the valuation date. Premiums that are expected to become due in the future should be incorporated in the cash-flow projection.

Recoverables

4.34 The valuation of the best estimate must be performed on a gross basis, without deducting amounts recoverable from eligible reinsurance and other eligible risk mitigation instruments.

4.35 Amounts recoverable from reinsurance contracts should be interpreted as the net amount recoverable from eligible reinsurance contracts, including reinsurance premiums, claim recoveries and other related cash-flows arising from the reinsurance arrangement.

4.36 The amounts recoverable from eligible reinsurance contracts and other eligible risk mitigation instruments must be calculated separately. Where the risk mitigation instruments involve arrangements with a Special Purpose Vehicle (SPV), the amounts recoverable from the SPV should not exceed the value of the assets an insurer could recover from the SPV.

4.37 When non-life insurers are performing this calculation, cash-flows relating to:

4.37.1 Provisions for claims outstanding should include the compensation payments relating to the claims accounted for in the gross provisions for claims outstanding of the insurer ceding risks; and

4.37.2 Premium provisions should include all other payments.

4.38 Payments in relation to settled insurance claims should not be included in the amounts recoverable from reinsurance contracts and other risk mitigation instruments. For example, debtors and creditors that relate to settled claims of policyholders or beneficiaries should not be included in the amounts recoverable from reinsurance contracts and special purpose vehicles. Recoverables may arise from reinsurance contracts and other risk mitigation instruments. The valuation of the best estimate must be performed on a gross basis, without deducting amounts recoverable from eligible reinsurance and other eligible risk mitigation instruments.

4.39 Recoverables from reinsurance arrangements that are not eligible as risk mitigation instruments must be excluded from the valuation of technical provisions.

4.40 Payments in relation to settled insurance claims should not be included in the amounts recoverable from reinsurance contracts and other risk mitigation instruments e.g., debtors and creditors that relate to settled claims of policyholders or beneficiaries should not be included in the amounts recoverable from reinsurance contracts and special purpose vehicles.

4.41 Expenses that the insurer incurs in relation to the management and administration of reinsurance contracts and other risk mitigation instruments must be included in the best estimate, calculated on a gross basis, without deducting amounts recoverable from reinsurance contracts and other risk mitigation instruments. No allowance for expenses relating to the internal processes should be made in the recoverables. Expenses involved include brokerage fees, tax, commission, claims audit expenses and general overhead costs. Actuarial judgement to be used also as these expenses differ from company to company.[1]

Discount Rates

i. Interim Approach – Proxy rates

4.42 In the interim, insurers and reinsurers shall apply the prescribed proxy discount rates provided by the Commission, until such a time where an active bond market will exist in the Zimbabwean market. The proxy rates shall be used to discount the insurer's technical provisions. See **TS 3 A2**.

ii. Long-term Approach – Government Bond Yield

4.43 In the presence of an active bond market in Zimbabwe, an insurer shall apply the government bond yield curve as the default for the risk-free term structure of interest rates.

4.44 The risk-free term structure shall be used to discount the insurer's technical provisions.

[1] Allowing for expenses incurred on reinsurance administration should be done on claims reserves. This provision in other words is an adjustment of claims reserves for Claims Handling Expenses. That is the best estimate IBNR or OCR provisions should take into account claims associated expenses. Refer to 4.41 of TS3.

4.45 In cases where the liability duration is longer than the term structure provided by the government bond yield curve, the insurer shall:

4.45.1 Use extrapolation techniques (see **TS 3 A2**) to determine the risk-free term structure; or

4.45.2 Assume that the yield curve shall remain flat from the latest term in the yield curve up to the point all liabilities expire.

Risk Margin

4.46 The insurer's technical provisions shall consist of the best estimate liability and the risk margin.

4.47 The purpose of the risk margin shall be to increase the insurer's technical provisions to the amount that would be paid by another insurer in order for that other insurer to take on the best estimate liability i.e., the risk margin is the part of the technical provisions that ensures that the value of the technical provisions is equivalent to the amount that another insurer would be expected to be paid to take over and meet the insurance obligations of the insurer.

4.48 The risk margin shall be used to increase the insurer's technical provisions to the amount that reflects the risk that the actual experience deviates from the best estimate assumptions.

4.49 Insurers and reinsurers under ZICARP should determine their risk margins using the approach in **TS 3 A3**.

5. Non-life Valuation Guidelines for Technical Liabilities

General Requirements

5.1 The general requirements when valuing technical provisions for non-life insurance companies are as follows:

5.1.1 Technical provisions for non-life insurance must be valued and disclosed for each line of business/proposed segmentations for nonlife insurance obligations under section **TS 3 A1** of this TS.

5.1.2 Proportional reinsurance must be treated as direct insurance by the reinsurer. The same principles used by the insurer must be mirrored on the reinsurer's reserving calculations;

5.1.3 Technical provisions for non-proportional reinsurance must be valued and disclosed for each proposed risk segmentation for nonproportional reinsurance under section **TS 3 A1** of this TS;

5.1.4 Insurance companies must use the principle of substance over form in treatment of contracts. Thus, non-life insurance liabilities that are similar in nature and characteristics to life insurance liabilities must be valued using life insurance principles. Additionally, non-life insurance companies must disclose separately non-life insurance liabilities that are valued using life insurance principles.

Methodology for Valuing Best Estimate

5.2 The best estimate refers to the **probability-weighted average** of an insurer's future cash-flows stemming from its insurance business, taking into account the time value of money and all possible scenarios of future potential outcomes. Although in principle, all possible scenarios should be considered, it may not be possible to explicitly incorporate all possible scenarios in the valuation. Consequently, an insurer or reinsurer may adopt a method that implicitly allows for all possible scenarios, such as the use of the chain-ladder technique (see IFRS 17, paragraph 33).

5.3 Non-life insurance companies must use actuarial and statistical techniques when valuing the best estimate, with the techniques adopted appropriately reflecting the risks which affect the cash-flows. Additionally, the approach used should take into account all factors that may materially affect the future claims experience. This may include simulation methods, deterministic techniques and analytical techniques. These methods will require non-life insurance companies to use claims data on a year of occurrence basis or an underwriting year basis for the run-off triangles.

5.4 Non-life insurance companies should provide a detailed description and justification of the actuarial method that they would have used to calculate best estimate provisions. Additionally, goodness of fit tests should be applied to all methods considered and only those methods that provide a good fit should be used. In circumstances where there are multi-treaties technical provisions can be segmented in a way approved by the actuary.

5.5 The methodology used to calculate technical provisions should be proportionate to the nature, scale and complexity of the risk supported by the insurer.

5.6 The calculation of best estimated provisions, where possible, should be done using homogeneous risk groupings and actuarial best practice. The results of the best estimate calculation must be reported using the risk segmentations for non-life insurance as presented under section **TS 3 A1** of this guideline. Insurers should adopt the approach that most appropriately captures the material characteristics of their risk profile, while satisfying the principal of proportionality.

5.7 For non-life insurers, a separate valuation of the best estimate must be carried out for provisions for claims outstanding and premium provisions. However, if such separation is not possible, non-life insurance companies may value these provisions together.

Valuation of Premium Provisions

5.8 Premium provisions replace the current Unearned Premium Reserves (UPR) and Unexpired Risk Reserves (URR).

5.9 Premium provision can be described as the expected present value of future cash inflows and outflows that will arise from the unexpired portion of the business that non-life insurance company is obliged to at the valuation date.

5.10 The cash-flow projection for the valuation should comprise of all future claim payments and claims administration/management expenses arising from these claim events; ongoing administration/management expenses of the existing policies and expected future premiums stemming from existing policies.

5.11 In some circumstances, the cash in-flows related to the premium provisions may exceed the cash outflows, leading to a negative best estimate. Reserves under ZICARP are not zeroised, the best estimate is taken as is. The valuation should take account of the time value of money where risks in the remaining period would give rise to claims settlements in the future.

Additionally, non-life insurance companies should refer to the contract boundaries of their policies when calculating premium provision. Only cashflows that fall within the contract boundary of the existing policies should be used to calculate premium provisions.

5.12 The valuation of premium provisions must also take account of future policyholder behaviour such as the likelihood of policy lapse during the remaining period.

Valuation of Claim Reserves

5.13 The cash-flow projections with respect to the best estimate for provisions for claims outstanding relate to claim events that have occurred before the valuation date, regardless of whether the claims arising from these events have been reported or not (i.e., all incurred but not settled claims).

5.14 The cash-flow projections for this component of the best estimate must comprise all future claim payments as well as claims administration/management expenses arising from the claim events. For clarity, all allocated and unallocated administration/management expenses associated with the claim events must be included in the cashflow projections for these provisions.

5.15 The reserves in respect of outstanding claims incurred and reported by the insurer shall be determined on the best estimate basis by using the case estimate method, the average cost per claim method or any other methods recognised by the Commission.

5.16 Insurance and reinsurance undertakings should not include:

5.16.1 The incurred but not reported provision (IBNR); and

5.16.2 Unallocated loss adjustment expenses (ULAE) in the calculation of the outstanding reported claims provision, which represents the component of the claims provision where events giving rise to the claim have been notified to the insurer.

5.17 The insurer's reserves in respect of incurred but not reported claims shall be valued and determined on the best estimate basis in accordance with the risk nature, risk distribution and experiential data of the insurance lines.

5.18 The methods to be adopted by the insurer for the valuation of the claim reserves shall depend on:

5.18.1 The particular characteristics of the class of business;

5.18.2 The reliability of the available data;

5.18.3 The past experience of the insurer and the industry;

5.18.4 The robustness of the valuation models; and

5.18.5 Consideration of materiality.

5.19 Where actuarial techniques (e.g., chain ladder techniques) are used to estimate incurred but not reported provision (IBNR), insurance and reinsurance undertakings should pay specific consideration to whether the assumptions behind the technique hold, or whether adjustments to development patterns are required to appropriately reflect the likely future development.

5.20 The value of the insurer's claim reserves shall include an amount in respect of the anticipated claim adjustment expenses.

5.21 When determining claim reserves, the insurer shall conduct a test on the adequacy of the reserves.

5.22 Claims adequacy test should be done where the insurer evaluates whether the reserves were sufficient to meet the claims experienced. The insurer should do a back-testing process to see if it was able to meet its insurance liabilities by comparing the actual experience against the assumed. The process of claims adequacy test can be broken down as follows:

5.22.1 Timing - Testing is required at each reporting date.

5.22.2 Type of testing - The test is a comparison of the carrying amount of insurance liabilities less related Deferred Acquisition Costs (DAC) and related intangibles to current estimates of future cash-flows under insurance contracts.

5.22.3 Recognition - If the net carrying amount is deficient, the entire deficiency is recognised in profit or loss.

5.23 Where the insurer's claims reserves are determined to be inadequate after conducting a test of the adequacy of the reserves, the insurer shall determine the claims deficiency reserves margin and load the margin to the reserve.

5.24 An insurer shall determine and disclose the value of its claims reserves for each class of business as per prescribed sublines in **TS 3 A1**.

6. Life Assurance Valuation Guidelines for Technical Liabilities

6.1 The insurer's technical provisions for life insurance business shall be composed of:

6.1.1 Best Estimate Liability; and

6.1.2 The risk margins.

6.2 Proportional reinsurance must be treated as direct insurance by the reinsurer. The same principles used by the insurer must be mirrored on the reinsurer's reserving calculations;

6.3 Technical provisions for non-proportional reinsurance must be valued and disclosed for each proposed risk segmentation for non-proportional reinsurance[2] under section **TS 3 A1** of this TS.

6.4 Insurance companies must use the principle of substance over form in treatment of contracts. Thus, non-life insurance liabilities that are similar in nature and characteristics to life insurance liabilities must be valued using life insurance principles. Additionally, non-life insurance companies must disclose separately non-life insurance liabilities that are valued using life insurance principles.

6.5 In addition to the general valuation guidelines (paragraphs 4.1 to 4.49), life insurers shall also consider the following business-specific guidelines.

[2] Reinsurance contracts must be valued in accordance to the requirements of TS3. Where multi-line contracts are issued, the company should determine a suitable approach that results in best estimate amount for the reinsurance recovery being determined. Additionally, the level at which this calculation should be aggregated and the approach to disaggregating the recovery amount to the various business lines depends on the data that is available to the actuary.

Options and Guarantees Embedded in Contracts

6.6 An insurer shall be required to identify all contractual options and financial guarantees embedded in the insurer's contracts when the insurer calculates the insurer's best estimate liability.

6.7 For each type of contractual option and guarantee, insurers must identify the risk drivers which have the potential to materially affect the current intrinsic value of the option or guarantee under a range of scenarios.

6.8 For contractual options, insurers must identify the risk drivers which have the potential to materially affect the frequency of option take-up rates under a range of scenarios.

6.9 The best estimate of contractual options and guarantees must capture the intrinsic value, timing and uncertainty of cash-flows, taking into account the likelihood and severity of outcomes from multiple scenarios combining the relevant risk drivers.

6.10 The best estimate of contractual options and financial guarantees shall:

6.10.1 Capture the uncertainty of cash-flows; and

6.10.2 Take into account the likelihood and severity of outcomes from multiple scenarios combining the relevant risk drivers.

6.11 An insurer shall value the best estimate of contractual options and financial guarantees by using one or more of the following methods:

6.11.1 A stochastic approach using a market-consistent asset model including both a closed-form and stochastic simulation approach;

6.11.2 Deterministic projections with attributed probabilities; and/or

6.11.3 A deterministic valuation based on expected cash-flows in cases where the valuation method delivers a market consistent valuation of the best estimate liability.

6.12 The approach used to value the best estimate of contractual options and guarantees should be proportionate to the nature, scale and complexity of the insurer's business.

6.13 If a stochastic simulation approach is used to value the best estimate of contractual options and guarantees, the approach should consist of an appropriate market-consistent asset model for projections of asset prices and returns. It should also comprise a dynamic model that incorporates the corresponding value of liabilities and the impact of any future management actions.

6.14 If a deterministic approach is used, a range of scenarios appropriate to both valuing the contractual options or guarantees and the underlying asset mix, together with the associated probability of occurrence, should be set. The probabilities of occurrence should be weighted towards adverse scenarios to reflect market pricing for risk and be sufficiently numerous to capture a wide range of possible outcomes.

6.15 Where the valuation of contractual options and guarantees is not performed on a policy-by-policy basis, the segmentation applied should not distort the value of the best estimate.

Discretionary Participation or With-Profit Contracts

6.16 Insurers that have policies with discretionary participation features must take into account future discretionary benefits which are expected to be made to policyholders when valuing technical provisions.

6.17 An insurer shall take into account future discretionary benefits which are expected to be made, whether or not the payments are contractually guaranteed in calculating the best estimate of the discretionary participation business.

6.18 The distribution of future discretionary benefits shall be a management action and assumptions about the distribution shall be objective, realistic and verifiable, and shall, in particular, take the relevant and material characteristics of the mechanism for their distribution into account.

6.19 The assumptions on the future returns of discretionary participation business should be consistent with the relevant risk-free interest term structure including where a risk-neutral approach for the valuation is used. Where a risk-neutral approach to the valuation of assets is used, assumptions on returns of future investments underlying the valuation of discretionary benefits should not exceed the level given by the forward rates derived from the risk-free interest rate term structure.

6.20 Where future discretionary benefits depend on the assets held by the insurer, the valuation of the best estimate should be based on the assets held by the insurer at the valuation date. Future changes to asset allocation should be treated as a separate future management action.

Methodology for Valuing the Best Estimate

6.21 The best estimate refers to the probability-weighted average of an insurer's future cash-flows stemming from its insurance business, taking into account the time value of money and all possible scenarios of future potential outcomes. (See IFRS 17, paragraph 33)

6.22 However, it may not be possible to explicitly incorporate all possible scenarios in the valuation and therefore, an undertaking may adopt a method that implicitly allows for all possible scenarios, such as the use of closed-form solutions.

6.23 Insurers must use actuarial and statistical techniques when valuing the best estimate, with the techniques adopted appropriately reflecting the risks which affect the cash-flows. This may include simulation methods, deterministic techniques and analytical techniques.

6.24 An insurer shall use actuarial and statistical methods in calculating the technical provisions that shall be proportionate to the nature, scale and complexity of the risk supported by the insurer.

6.25 The valuation of the best estimate should involve cash-flow projections that take into account the timing and uncertainty of future cash-flows.

6.26 The valuation should consider the variability of cash flows and ensure that the best estimate represents the mean of the distribution of potential outcomes.

6.26.1 The starting point for an estimate of the cash flows is a range of scenarios that reflects the full range of possible outcomes. Each scenario specifies the amount and timing of the cash flows for a particular outcome, and the estimated probability of that outcome. The cash flows from each scenario are discounted and weighted by the estimated probability of that outcome to derive an expected present value. Consequently, the objective is not to develop a most likely outcome, or a more-likely-than-not outcome, for future cash flows.

6.26.2 In practice, developing explicit scenarios is unnecessary if the resulting estimate is consistent with the measurement objective of considering all reasonable and supportable information available without undue cost or effort when determining the mean. In some cases, relatively simple modelling may give an answer within an acceptable range of precision, without the need for many detailed simulations.

6.26.3 Similarly, in some cases, the cash flows may be driven by complex underlying factors and may respond in a non-linear fashion to changes in economic conditions. This may happen if, for example, the cash flows reflect a series of interrelated options that are implicit or explicit. In such cases, more sophisticated stochastic modelling is likely to be necessary to satisfy the measurement objective.

6.27 Deterministic models are mainly used for this but stochastic models that can reflect real-world economic scenarios that provide a range of possible outcomes to model variability of cashflows can also be used. They allow for the possibility of high volatility and a sequence of bad returns in the early years. Markov chain processes can be used to model cashflows. These use matrix algebra in excel to forecast outcomes, given a starting point and probabilities that describe the chance of transitioning from one state to another. Another example is the use of the Mack model which uses past claims data to derive estimates of the mean and variance of the total ultimate claims arising from each origin period. It makes no assumption or prediction about the precise distributions involved and so is described as a distribution-free model.

6.28 The valuation of life insurance obligations should generally be based on a policy-by-policy approach i.e., the cash-flow projection by an insurer shall be based on a policy-by-policy approach but reasonable actuarial methods and approximations may be used.

6.29 In certain circumstances, the best estimate may be negative (e.g., for some individual policies). Reserves under ZICARP are not zeroised. The valuation should take account of the Time Value of Money where risks in the remaining period would give rise to claims settlements in the future.

6.30 No implicit or explicit surrender value floor should be assumed for the market-consistent value of liabilities of a life insurance policy. This means that if the sum of the best estimate and the risk margin for a policy is lower than the surrender value of the policy, the value of the insurance liabilities should not increase to the surrender value of the policy.

6.31 Cash-flow projections must reflect expected realistic future demographic, legal, medical, technological, social or economic developments.

6.32 Cash-flow projections must incorporate appropriate assumptions for future inflation, including appropriate recognition of the type of inflation to which particular cash flows are exposed (e.g., consumer price index or salary inflation).

6.33 Insurers may use premium adjustment clauses to offset the effects of claims inflation for certain policies, however, insurers must be satisfied that any offset does not undervalue the best estimate or underestimate the risks involved.

6.34 The projection horizon used must cover the cash in-flows and out-flows within the contract boundary, required to settle insurance obligations of existing policies. Determining the lifetime of insurance obligations must be based on up-to-date and credible information, as well as realistic assumptions about when existing insurance obligations will be discharged, cancelled or expired.

6.35 Insurers must not recognize, as an asset or a liability, any amounts relating to expected premiums that are outside the contract boundary. An insurance policy should be derecognized as an existing contract only when the obligation specified in the policy expires or has been discharged or cancelled. The best estimate liability must be determined as the discounted value of projected cash flows up to the contract boundary.

6.36 The best estimate must be valued separately for obligations in different currencies.

6.37 Where an insurer is determining the value of the insurer's technical provisions, the insurer shall use a market consistent approach. Mark-to-market approach is the default approach and where it is not possible then the mark-to-model approach must be used.

6.38 Market-consistent valuation almost invariably involves the following actuarial practice:

6.38.1 Use a risk-neutral stochastic model to project the cashflows of the asset or liability that is, liabilities from long-term savings products such as with-profits policies. Note the model may not need to be implemented using a simulation model but an analytical valuation of the instrument could be derived from the specified stochastic process.

6.38.2 The projected risk-neutral cash-flows of the instrument are discounted using risk-free interest rate term structures. The starting interest rate term structure could be observed from government bonds, but interest rate swaps are used in some circumstances. The long-term nature of the liabilities means that the part of the yield curve that can be observed from market prices usually needs to be extrapolated to longer maturities. There may also be an adjustment to the observable risk-free term structure that is made to allow for the illiquidity of the insurance liabilities.

6.38.3 The stochastic model's volatility assumptions are calibrated to market prices that are sensitive to volatility such as equity options and interest rate swaptions. Again, the available market prices may not provide a term structure of volatility assumptions that is sufficiently long-term to be directly applicable to the insurance cash flows that are being valued. So again, some extrapolation method is required.

6.38.4 In order to demonstrate that the model is not market-inconsistent, a series of quantitative tests of the model output would be produced that show that the model produces valuations for elementary assets such as bonds that are consistent with the observed or assumed market prices.

6.39 In determining the best estimate liability, the insurer shall consider the following cash inflows:

6.39.1 Future premiums; and

6.39.2 Future fund management charges or fees income.

6.40 In determining the best estimate liability an insurer shall consider the following policyholder benefit cash outflows:

6.40.1 Death benefits;

6.40.2 Critical illness and disability benefits;

6.40.3 Surrender benefits;

6.40.4 Partial and full maturity benefits;

6.40.5 Annuity payments; and

6.40.6 Profit share commission payments.

6.41 In determining the best estimate liability, the insurer shall consider the following expense cash outflows:

6.41.1 Administrative expenses;

6.41.2 Investment management expenses;

6.41.3 Claims management or handling expenses;

6.41.4 Direct and override commissions which are expected to be incurred in the future;

6.41.5 Overheads expenses;

6.41.6 Premium levy and policyholder compensation levy; and

6.41.7 Overhead expenses which shall include those related to general management and service functions that are not directly involved in new business or policy maintenance.

6.42 Under certain conditions, however, simplified actuarial techniques and approximations may be used to group policies when projecting future cash flows for these policies. The conditions for consideration include:

6.42.1 The grouping of policies does not misrepresent the underlying risk or significantly misstate the costs of satisfying the relevant insurance obligations;

6.42.2 The grouping of policies does not distort the valuation of technical provisions; and

6.42.3 Sufficient validation has been performed to verify that the grouping of life policies does not lead to the loss of any significant attributes of the portfolio being valued. Special attention should be given to the number of guaranteed benefits and any possible restrictions (legislative or otherwise) for an insurer to treat different groups of policyholders fairly.

6.43 The Actuary may adopt any other valuation method for the valuation of technical provisions, provided it shall not result in a value lower than the value obtained using the best estimate assumptions and risk margin.

7. TS 3 A1: Risk Classifications and Contract Boundaries

Life Assurance Segmentation

Main Class	Sub-Class 1	Sub-Class 2	Sub-Class 3	Proposed Contract Boundary
Annuities	Level Annuities	Level annuities	Level annuities	**Guaranteed Deferred annuities** If a contract provides a savings/accumulation phase and at maturity the accumulated benefits/savings may be paid out as a lump sum or as an annuity, for which a guaranteed annuity rate is provided; the insurer should only include the premiums up until the maturity date of the deferred annuity contract. The cash flows of the second contract (the annuity contract) should not be taken into account. However, the insurer should also estimate the cost of issuing a guarantee or option with the first contract (i.e., there would be a need to value the guaranteed annuity option as it is an inherent part of the first contract). If a contract provides for an accumulation phase which is followed by an annuity phase and in substance, the terms of the contract are such that this represents a single contract, which has both accumulation and annuity phases, then all cash flows for the whole contract life should be included.
	Growth Annuities	With discretionary participating features	With discretionary participating features	
		Market Linked	Market Linked	
Risk Products	Term Assurance	Individual	Individual	**Individual Policies** The following is the approach to contract boundaries for individual risk policies: i. Where the pricing of the premiums takes into account risks that relate to future periods, the boundary should generally be the same as the contract term. Where insurers have the ability to implement certain management actions, such as changing future premium rates after an initial guaranteed term, such actions are within the contract boundary. Allowance should therefore be made for the management actions that the insurer could reasonably be expected to implement and allow appropriately for expected policyholder behaviour.
		Group	Group	
	Endowment	Individual	Without discretionary participating features	

			With discretionary Participating features	ii. If the insurer can only price and re-price products on a portfolio level, we assume that the insurer's unilateral right to change policy conditions to fully reflect the risk inherent in the specific policy is limited and therefore a longer contract boundary should apply. A shorter boundary can only apply if the insurer should have the right to set premium rates for existing policies that are the same as that for new policyholders with the same risk profile and also has the right to re-underwrite the policies using the same rules as those applied to new policies. iii. Contact boundaries should not be longer than the contractual end of the policy. **Group Policies:** The contract boundary should be the next review date.
		Group	Without discretionary participating features	
			With discretionary Participating features	
	Pure Endowment	Individual	Without discretionary participating features	
			With discretionary Participating features	
		Group	Without discretionary participating features	

			With discretionary Participating features	
		Whole life	Without discretionary participating features	
	Whole Life	Whole life	With discretionary Participating features	
			Without discretionary participating features	
		Individual	With discretionary Participating features	
	Funeral	Group	Without discretionary participating features	

			With discretionary Participating features	
	Credit Life Assurance	Individual	Individual	
		Group	Group	
	Group Life Assurance	Group Life Assurance	Group Life Assurance	The contract boundary should be the point where the insurer has the unilateral right to change policy conditions on a scheme level to fully reflect the risk inherent in the policy. Where the premium rates are reviewed at a scheme level on an annual basis, the contract boundary should be the next review date (unless rates are guaranteed for a longer period).
Investment Contract	Individual	Guaranteed	Guaranteed	**Contract boundaries for linked investment policies** The value of linked policies should be equal to the number of units multiplied by the unit price i.e., insurers should apply a zero-contract boundary for linked investment policies. **Contract boundaries for investment policies with no financial guarantees** The contract boundary should be the point where the insurer has the unilateral right to change policy conditions on a contract level 2 to fully reflect the risk inherent in the policy. This boundary should not be longer than the contractual end of the policy. At the contract boundary, an allowance should be made for the full projected account balance to be paid to the policyholder. If the contract is open-ended and the insurer does not have the unilateral right to change policy conditions or terminate the policy, then a long contract boundary should be assumed with persistent assumptions dictating the run-off of the business.
		Linked	Linked	
		Market Linked	Market Linked	
		With discretionary participating features	With discretionary participating features	
	Fund Business	Guaranteed	Guaranteed	
		Linked	Linked	

		Market Linked	Market Linked	**Contract boundaries for investment policies with financial guarantees:** If an investment policy has a financial guarantee, the insurer should use a contract boundary that is the greater of: a. the point where the insurer has the unilateral right to change policy conditions on a contract level to fully reflect the risk inherent in the policy; and b. the point where the financial guarantee ends. The limit is that the boundary should not be longer than the contractual end of the investment policy. At the contract boundary, an allowance should be made for the full projected account balance to be paid to the policyholder.
		With discretionary participating features	With discretionary participating features	
				If an investment policy is open-ended and the insurer does not have the unilateral right to change policy conditions or terminate the policy, a long contract boundary should be assumed with persistent assumptions dictating the run-off of the business.
		Proportional	Treaty	The contract boundaries for life reassurance should be determined as the point: i. where the reinsurer has the unilateral right to change the policy conditions to fully reflect the risk inherent in the contract or to terminate the contract unless the reinsurer can compel the insurer (policyholder) to pay the premiums for future obligations beyond this point.
			Facultative	
			Treaty	

Reinsurance	Life Reassurers	Non-proportional	Facultative	ii. If the terms of a treaty allow the reinsurer to increase rates at a review point, and where at such a point the cedant insurer has the option to either terminate or continue with the treaty under the new terms then the following will apply: a. if it is expected that the terms will remain unchanged, the contract boundary shall be equal to the contract term as future premiums can be compelled. b. if the treaty is profitable to the reinsurer, the contract boundary shall be equal to the underlying contract term. c. if the treaty is unprofitable and it is the best estimate view of the reinsurer that rates will be reviewed at the next review date, then the next premium review date should be used as the contract boundary. If for such a contract the best estimate view of the reinsurer is that rates will not be increased, then the contract boundary shall be equal to the underlying contract terms. d. The contract boundary for reinsurance of group life business is generally the next renewal date. Where the contract includes an automatic renewal clause, and the cancellation date has passed, then the contract boundary is the renewal date plus the period for which the new rate is being guaranteed. e. Profit commission should be included using an economic view over the contract term.

Non-life insurance segmentations

Main Class	Sub-class 1	Sub-class 2
Pure Liability	Personal Liability	
	Employers Liability	Workmen's compensation
		Domestic worker's compensation
Property Damage (with liability)	Engineering	
	Property	Personal lines
		Commercial lines
	Farming*	Crop
		Livestock
		Equipment
		Other
	Motor*	Motor private – Private Vehicles Comprehensive
		Motor private – Private Vehicles Third Party
		Motor private – Private Vehicles Third Party and Fire
		Motor Commercial – Commercial Vehicles
		Motor Commercial – Commercial PSV
	Fire	Domestic Risks
		Industrial and Commercial risks
	Aviation	Property
		Liability
	Marine	Marine Hull
		Marine Cargo
Financial Loss	Bonds	
	Hire Purchase	
Fixed Benefits		Personal Accident and Sickness

	Personal Accident Insurance	Health/Medical Expenses Insurance (where separate policies are issued)
	Micro-insurance	
	Travel insurance*	
	Legal expense insurance*	
	Terrorism	Motor
		Property
		Engineering
		Other
	Credit and guarantee	
	Miscellaneous Accident	
	Cell Captive	Pure captives
		Group and/or association captives
		Rental captives (Cell Insurance)
		Diversified captives
Reinsurance	Proportional reinsurance	
	Non-proportional reinsurance	Marine, aviation and transport reinsurance
		Property reinsurance
		Liability reinsurance

8. TS 3 A2: Risk-Free Term Structure under ZICARP

8.1 The framework under ZICARP recommends **2 methods**:

8.1.1 Interim Method

8.1.2 Long-term Method.

Interim Method – Proxy Rates

8.2 In the absence of an active bond market that can consistently provide yield rates, this approach relies on the Commission or the Reserve Bank of Zimbabwe to provide proxy discount rates[3] for varying levels of maturity.

8.3 The interim method proposes a simplified methodology, where proxy discount rates for each type of insurance business are published through periodic[4] publications by the Commission or the Reserve Bank of Zimbabwe that insurers use in discounting.

Long-term Method – Extrapolation of Government Bond Yield Curve

8.4 Once bond market data becomes available in Zimbabwe, this method proposes that insurers shall apply the government bond yield curve as the default for the risk-free term structure of interest rates.

8.5 For most term structures, there is sufficient liquidity up to a certain maturity. Beyond this point, the term structure needs to be extrapolated when necessary.

8.6 Extrapolation and/or interpolation techniques can then be implemented using appropriate extrapolation methods like the Smith-Wilson model.

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[3] The interim proxy risk-free discount rates will be based on estimates of the Ultimate Forward Rates. See the section on estimation of interim Ultimate Forward Rates by IPEC under ZICARP

[4] The requirement is that IPEC or RBZ publishes risk-free discount rates at intervals of at most 12 months.

9. TS 3 A3: Risk Margins on Technical Provisions under ZICARP

9.1 Insurance companies under ZICARP should determine their risk margins using the Cost of Capital method for the following reasons:

9.1.1 It captures the market value of insurance liabilities;

9.1.2 It takes into account the skewness in the distributions (claims distribution functions) of insurance liabilities;

9.1.3 It allows for the time value of money or allows for discounting of future cash flows;

9.1.4 It aligns with other international risk-based regulatory frameworks like Solvency II and SAM; and

9.1.5 It is relatively straightforward and transparent, therefore more understandable to clients, regulators and other stakeholders.

9.2 Besides the Cost of Capital method, insurers can for practical expediency elect to apply the Risk Adjustment used in IFRS 17 reporting as the objectives broadly overlap with the above ZICARP principles.

Assumptions

9.3 The risk margin calculation is based on the following ten (10) transfer scenario or assumptions:

9.3.1 The whole portfolio of insurance and reinsurance obligations of the insurance or reinsurer that calculates the risk margin (original insurer) is taken over by another insurance or reinsurer (reference insurer);

9.3.2 The transfer of insurance and reinsurance obligations includes any reinsurance contracts and arrangements with special purpose vehicles relating to these obligations;

9.3.3 The reference insurer does not have any insurance or reinsurance obligations and any own funds before the transfer takes place;

9.3.4 After the transfer the reference insurer raises eligible own funds equal to the SCR necessary to support the insurance and reinsurance obligations over the lifetime thereof;

9.3.5 After the transfer the reference insurer has assets to cover its SCR and the technical provisions net of the amounts recoverable from reinsurance contracts and special purpose vehicles;

9.3.6 The assets should be selected in such a way that they minimize the SCR for market risk that the reference insurer is exposed to;

9.3.7 The SCR of the reference insurer captures underwriting risk, unavoidable market risk, credit risk and operational risk;

9.3.8 The loss-absorbing capacity of technical provisions in the reference insurer corresponds to the loss-absorbing capacity of technical provisions in the original insurer;

9.3.9 There is no loss-absorbing capacity of deferred taxes for the reference insurer; and

9.3.10 Without prejudice to the transfer scenario, the reference insurers will adopt the same future management actions as the original insurer.

9.4 The assumption in the calculation of the risk margin is that the reference insurer at time t = 0 (when the transfer takes place) will capitalise itself to the required level of eligible own funds (excess of assets over liabilities), i.e.

$EOF_{RU}(0) = 0)$

Where:

$EOF_{RU}(0)$ = the amount of eligible own funds raised by the reference insurer at time t = 0; and

$SCR_{RU}(0)$ = the SCR at time t = 0 as calculated for the reference insurer.

9.5 The transfer of (re)insurance obligations is assumed to take place immediately. The formula for coming up with the risk margin is as follows:

$$Risk\ Margin = CoC . \sum \frac{SCR_{RU}(t \qquad\qquad)}{(1 + r \quad)^{t+1}{}_{+1}}$$

$t \geq 0$ t

Where:

$SCR_{RU}(t)$ = the SCR for period t as calculated for the reference insurer,

r_t= the risk-free rate for maturity t; and

CoC= the Cost of Capital rate.

9.6 The risk-free rate r_t for the discounting of the future SCRs should not include an illiquidity premium.

9.7 The calculation of the risk margin above applies to all insurers (life and nonlife insurance undertakings).

9.8 It should be ensured that the assumptions made regarding the loss absorbing capacity of technical provisions to be taken into account in the SCR calculations are consistent with the assumptions made for the overall portfolio of the original insurer (i.e., the insurer reporting under ZICARP).

Treatment of Market Risk in the calculation of Risk Margins

9.9 With respect to market risk, only the unavoidable market risk should be taken into account in the risk margin in cases where it is significant.

9.10 With regards to non-life insurance obligations and short-term and mid-term life insurance obligations the unavoidable market risk can be considered to be nil. For long-term life insurance, there might be an unavoidable interest rate risk. It is not likely to be material if the duration of the insurer's whole portfolio does not exceed the duration of risk-free financial instruments available in financial markets for the currencies of the portfolio. The assessment of whether the unavoidable market risk is significant should take into account that it usually decreases over the lifetime of the portfolio.

Treatment of Counterparty Risk

9.11 For counterparty default risk, only the risk for ceded reinsurance should be taken into account in the risk margin.

Non-Life Insurance Obligations

9.12 With respect to non-life insurance, the risk margin should be attached to the overall net of reinsurance best estimate and will not be split between premium provisions and outstanding claims.

The Cost-of-Capital rate

9.13 The Cost-of-Capital rate is the annual rate to be applied to the capital requirement in each period and it is the spread over and above the risk-free rate.

9.14 Insurance companies under ZICARP should determine their Cost-of-Capital rate using acceptable corporate finance methods like Weighted Average Cost of Capital or Marginal Cost of Capital[5].

Level of granularity in the risk margin calculations

9.15 The risk margin should be calculated per line of business. A straightforward way to determine the margin per line of business is as follows: First, the risk margin is calculated for the whole business of the insurer, allowing for diversification between lines of business. In the second step, the margin is allocated to the lines of business.

[5] The Cost of Capital Rate under Solvency II and SAM is 6%. This rate has been maintained under the ZICARP framework.

9.16 The risk margin per line of business should take the diversification between lines of business into account. The sum of the risk margin per line of business should be equal to the risk margin for the whole business. The allocation of the risk margin to the lines of business should be done according to the contribution of the lines of business to the overall SCR during the lifetime of the business.

9.17 The contribution of a line of business is to be analysed by calculating the SCR under the assumption that the insurer's other business does not exist. Provided the relative sizes of the SCRs per line of business do not materially change over the lifetime of the business, insurers may execute the following formula:

$$Risk\ Margin_{lob} = \frac{SCR_{RU,lob}(0)}{\sum_{lob} SCR_{RU,lob}} \cdot COCM, \qquad (0)$$

Where:

$Risk\ Margin_{lob}$= risk margin allocated to the line of business (lob);

$SCR_{RU,lob}(0)$ = SCR, relating to unavoidable risks only, of the reference insurer for the line of business (lob) at *t=0*; and

$COCM$= risk margin for the whole business.

9.18 If a line of business consists of obligations where the technical provisions are calculated as a whole, the formula should assign a zero-risk margin to this

line of business. Because $SCR_{RU,lob}(0)$ of this line of business should be zero.

Simplifications on the calculation of risk margins for non-life insurance

9.19 Non-life insurance companies can calculate risk margins as a percentage of their net of reinsurance best estimate technical provisions by applying the following prescribed percentages:

Line of Business	Prescribed %
Motor	10%
Liability lines of business	20%
Other lines of business other than Motor and Liability	15%

Overall Simplification and Alignment with IFRS 17

9.20 Given that the new IFRS 17 has a requirement for the determination of a Risk Adjustment that has some broad commonalities with the Risk Margin under ZICARP, both Life and Non-Life insurers will have a choice to adopt the IFRS 17 Risk Adjustment as the ZICARP Risk Margin as a simplification.

10. TS 3 A4: Guidance and Simplification on Premium Provisions

Guidance on calculation of unearned premium reserve.

10.1 An undertaking shall determine and disclose the value of its unearned premium reserves for **each** class of business.

10.2 When determining unearned premium reserves, an insurer shall conduct a test of the adequacy of the reserves.

10.3 Where the unearned premium reserves are inadequate, the insurer shall determine the premium deficiency reserves.

10.4 The insurer shall be required to disclose the method it uses to determine its unearned premium reserve and all changes in the method shall have to be justified and be approved by the Commission.

10.5 The reserving method used by the insurer to determine the unearned premium reserves shall not be changed arbitrarily by the insurer.

10.6 An insurer shall, in determining the value of unearned premium reserves, apply the following methods:

a) “12ths” method (reserving on a monthly basis);
b) “24ths” method (reserving on a monthly basis);
c) “365ths” method (reserving on a daily basis);
d) “8ths” method (reserving on a quarterly basis); and
e) any other method as may be approved by the Commission.

10.7 A reinsurer may apply the “8ths” method when reserving on a quarterly basis.

10.8 The insurer shall calculate the reserve for the insurer's unexpired risks and this reserve for unexpired risks shall be calculated and maintained separately for each class of insurance.

Simplifications on calculating premium provisions

10.9 Non-life insurance companies can use the Expected Loss Based Proxy (ELBP) to estimate premium provisions. The ELBP can be used with the combined ratio that is primarily estimated from the insurance company's data.

10.10 The insurance company's combined ratio is defined as:

$$\frac{Incurred\ Claims_i + Expenses_i}{Earned\ Premiums_i}$$

Where:

$Incurred\ Claims_i$= Incurred Claims per line of business i;

$Expenses_i$= expenses per line of business i; and

$Earned\ Premiums_i$= earned premiums per line of business i.

10.11 Earned premiums, expenses and incurred claims should be defined and calculated consistently with the contract boundary requirements under section TS 3 A1 of this guideline and IFRS 17. Any prior year adjustments on earned premiums should be excluded from the calculation. The expenses should be those attributable to the premiums earned other than claims expenses. Incurred claims should exclude the run-off result, that is they should be the total for losses occurring in year x of the claims paid (including claims expenses) during the year x and the provisions established at the end of the year x.

10.12 A practical alternative to the calculation of loss ratio is to add expense ratio for year x and claims ratio for year x. In this case, they should be calculated as follows:

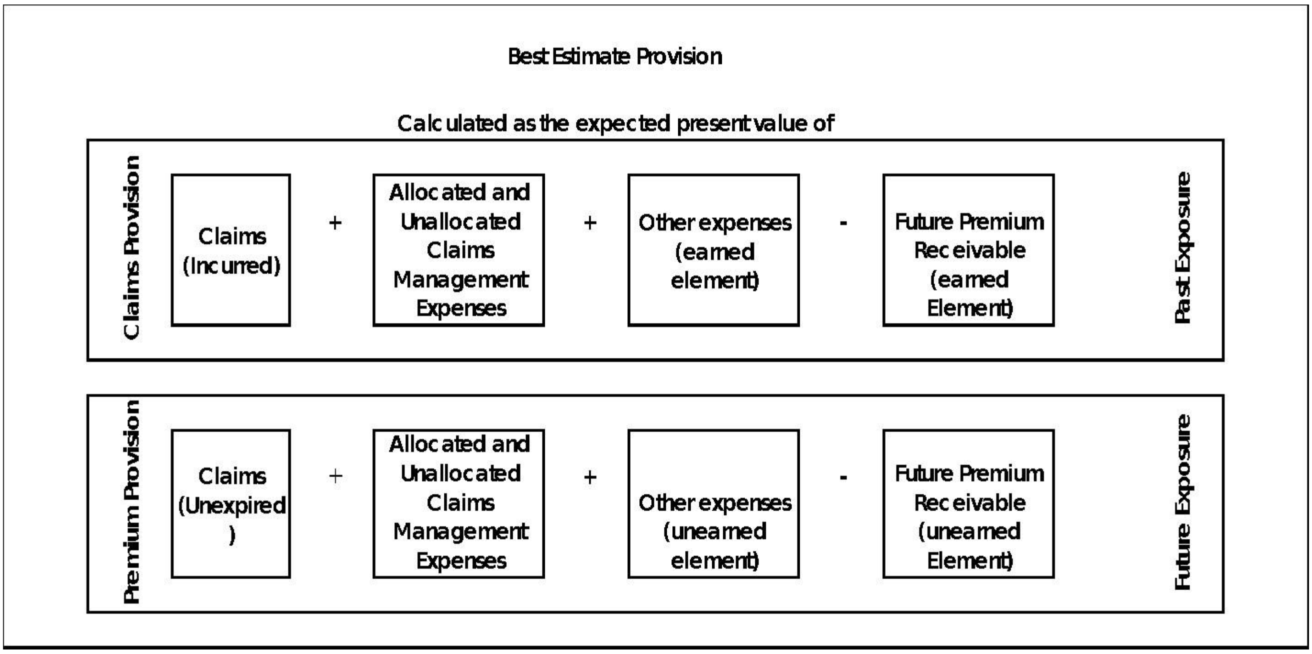
$$Expense\ ratio = \frac{Expenses\ exlcuding\ claim\ expenses}{written\ premiums}$$; and

$$Claims\ ratio = \frac{Incurred\ claims\ for\ line\ of\ business\ i}{earmed\ premiums\ for\ line\ of\ business\ i}$$

10.13 The expenses used in the calculation of the expense ratio are those that are attributable to the written premiums.

11. TS 3 A5: Relationship between claims and premium provisions

11.1 The diagram below shows the relationship between provision for outstanding claims and premium provision:



Adopted from: (Institute and Faculty of Actuaries General Insurance Reserving Oversight Committee's Working Paper of Solvency II Technical Provisions, August 2013)

12. Glossary

Reinsurance: Type of risk mitigation on the basis of an insurance contract between one insurer or pure reinsurer (the reinsurer) and another insurer or pure insurer (the cedant), to indemnify against losses, partially or fully, on one or more contracts issued by the cedant in exchange for a consideration (the premium).

Required economic capital: The total of assets measured at market-consistent value, internally required by an insurer above the market consistent value of obligations, in order to reduce the risk of not meeting the obligations to a defined risk measure, and within a defined time period (e.g., one year).

Risk margin: A generic term, representing the value of the deviation risk of the actual outcome compared with the best estimate, expressed in terms of a defined risk measure.

Technical provision: An insurer's contractual obligations/rights under an insurance contract.

Regulatory capital: The capital level representing the final threshold that triggers ultimate supervisory measures in the event that it is breached.